

ITI113 Project Proposal

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1. Problem Framing / Formulation

Cryptocurrency-related scams are an increasing global and local concern, as fraudsters exploit digital messaging platforms to reach victims quickly and at scale. In Singapore, this threat is particularly acute, with cryptocurrency-related scams accounting for about 20% of total scam losses in 2025¹. Despite growing awareness, users often lack timely and reliable support to assess whether crypto-related messages are fraudulent before taking action. To address this gap, the project frames cryptocurrency scam detection as a supervised binary text-classification problem, where the input is a crypto-related message and the output is a binary label indicating scam or legitimate. By automatically distinguishing scam messages from legitimate crypto-related communications, the project aims to provide early, decision-support assistance to users using a data-driven and interpretable machine learning (ML) approach.

While simple rule-based methods or heuristic approaches can be used as a baseline, they are vulnerable to changes in scam language. Baseline heuristics include keyword-based blacklists (e.g., ‘guaranteed returns’) and rule-based detection of wallet addresses or URLs. However, scammers frequently change their wording, making such systems difficult to maintain and resulting in reduced detection performance over time. In contrast, a supervised ML approach can capture broader textual signals and better adapt as scam tactics evolve. Lastly, consistent with recent research, this project focuses on interpretable, classical models as a proof-of-concept, while more advanced language models (e.g., using large language models (LLM)) are left as potential future work^{2,3,4}.

Model performance will be evaluated using:

- Accuracy
- Confusion Matrix
- Precision
- Recall
- F1 Score

¹ Singapore Police Force, 2025. *Annual Scam and Cybercrime Brief 2025*. [Online] Available at: <https://www.scamshield.gov.sg/archive/2025-scams-cybercrime-briefs/>

² Kim, H., Gim, M., Choi, D., Lee, H., Bae, S., Kim, M.-Y., & Kang, J. (2026a). *An LLM-based chain-of-response counter-scam system*. IJCAI 2026. <https://arxiv.org/abs/2606.01475>

³ Kim, H., Kim, C., Shin, S., & Kang, J. (2026b). *SCRIPTMIND: Crime script inference and cognitive evaluation for LLM-based social engineering scam detection system*. EACL 2026 Industry Track. <https://arxiv.org/abs/2601.13581>

⁴ Senol, A., Agrawal, G., & Liu, H. (2026). Domain knowledge-enhanced LLMs for fraud and concept drift detection. *Electronics*, 15(3), 534. <https://doi.org/10.3390/electronics15030534>

- ROC Curve/AUC
- Precision–Recall (PR) Curve/AUC

A successful outcome is defined by achieving a high recall ($\geq 85\%$) to minimise missed scam messages (false negatives), as undetected scams may expose users to financial loss or identity theft. At the same time, the model should maintain an acceptable precision ($\geq 75\%$) to reduce unnecessary false alarms on legitimate messages. Overall model performance will also be evaluated using an F1-score of at least 80% to balance precision and recall, and a ROC-AUC of at least 0.85 to demonstrate good discrimination between scam and legitimate messages.

2. Dataset Identification

Dataset Overview

- Dataset name: Crypto Scam Dataset
- Source/Link: <https://www.kaggle.com/datasets/theeyanyan/crypto-scam-dataset/data>
- Target dataset size: 10,000 records
- Data format: Structured tabular dataset (e.g., CSV)
- Target variable: label - scam or legitimate/legit

Dataset Schema

- id: Unique message identifier
- platform: Source platform (e.g., Telegram, X/Twitter, SMS, Email, Discord, Reddit)
- text: Message content
- label: Binary label (scam or legitimate)

The dataset consists of synthetically generated crypto-related messages sourced from Kaggle. Genuine scam data is inherently sensitive and rarely accessible due to privacy constraints, which means synthetic alternatives are often the most practical option available. The dataset was designed to provide balanced, linguistically diverse coverage of common scam typologies alongside everyday legitimate crypto communication. No real individuals, wallets, or platforms are involved, minimising privacy risk. Platform diversity is included to improve robustness at a proof-of-concept level.

The synthetic nature of the data may over-represent certain scam styles and under-represent others. The dataset is suitable for learning and demonstration purposes but not for deployment without further validation on real-world data.

3. Data Governance

Data governance focuses on data quality, bias, representativeness, and privacy. The synthetic dataset helps avoid personal data exposure but may introduce bias from the generation process. Exploratory data analysis (EDA) will be used to examine class imbalance, feature distribution, and platform representation. Random seeds and fixed train–test splits will be

used to ensure consistent experimental results across model iterations. These considerations will be documented to ensure responsible data use.

All trained models will also be version-controlled through GitHub and SageMaker model artefacts to ensure traceability and reproducibility. If a newer model performs worse, the previous version can be restored quickly.

4. Methods

The project uses a supervised ML binary classification approach with the message text as input features and the scam/legitimate label as the target variable. Term Frequency–Inverse Document Frequency (TF–IDF) is chosen for text representation because it is a well-established, interpretable technique for weighting important words in short messages, and is suitable for labelled datasets commonly used in classical scam-detection tasks.

Feature Engineering/Extraction:

The project extracts a set of engineered features that can be explored alongside the model.

- **Urgency indicators:** Keyword-based detection (e.g. time-pressure phrases), exclamation mark frequency, countdown phrases, and a composite urgency score
- **Contact and link characteristics:** Presence and count of wallet addresses, presence of phone numbers, email addresses, and hyperlinks
- **Structural characteristics:** Exclamation count, capitalisation patterns, and general numeric presence

Candidate models may include:

- Baseline heuristic/statistical model (e.g., keyword-based rules)
- Logistic Regression as an interpretable baseline
- Naive Bayes
- Support Vector Machine
- Tree-based models such as Decision Trees or Random Forests
- Neural Network (for comparison, subject to project scope)

Models will be compared using the defined success metrics, with attention to class imbalance, interpretability, and practical deployment considerations. The final model will be selected based on predictive performance, interpretability, computational efficiency, and suitability for deployment within the Streamlit application.

5. Deployment

Intended Use

The model will be deployed as a lightweight web application using Streamlit to demonstrate a real-world proof-of-concept. The application is intended as a decision-support tool for

members of the public who receive suspicious cryptocurrency-related messages via platforms such as Telegram, WhatsApp, Twitter/X, Discord, Reddit or Email.

Users can paste the message into the application, where it will be preprocessed and transformed into TF-IDF features before being passed to the trained machine learning model for prediction. The application will return:

- Risk category (Low / Medium / High)
- Estimated Scam Probability (e.g., 92%)
- Brief explanation of the result
- Safety advisory for the user

If the predicted risk is *medium* or *high*, the application will advise users to stop before taking any action, note down key information from the suspicious message (e.g., the platform, URL, phone number, email address, or sender details, where available), and verify the information through official channels (e.g., the ScamShield Helpline at 1799 or the relevant authorities). This information can also facilitate reporting through ScamShield or other relevant reporting channels.

Even when the predicted risk is *low*, the application will clearly state that the prediction does not guarantee legitimacy, and users should continue exercising caution before transferring funds or sharing sensitive information.

Sample display message to the user:

Predicted Risk Level: High-Risk

Estimated Scam Probability: 92%

Key features or keywords contributing to the prediction:

- Presence of urgent or time-sensitive language
- Cryptocurrency wallet address detected
- Claims of guaranteed or unusually high investment returns

Explanation:

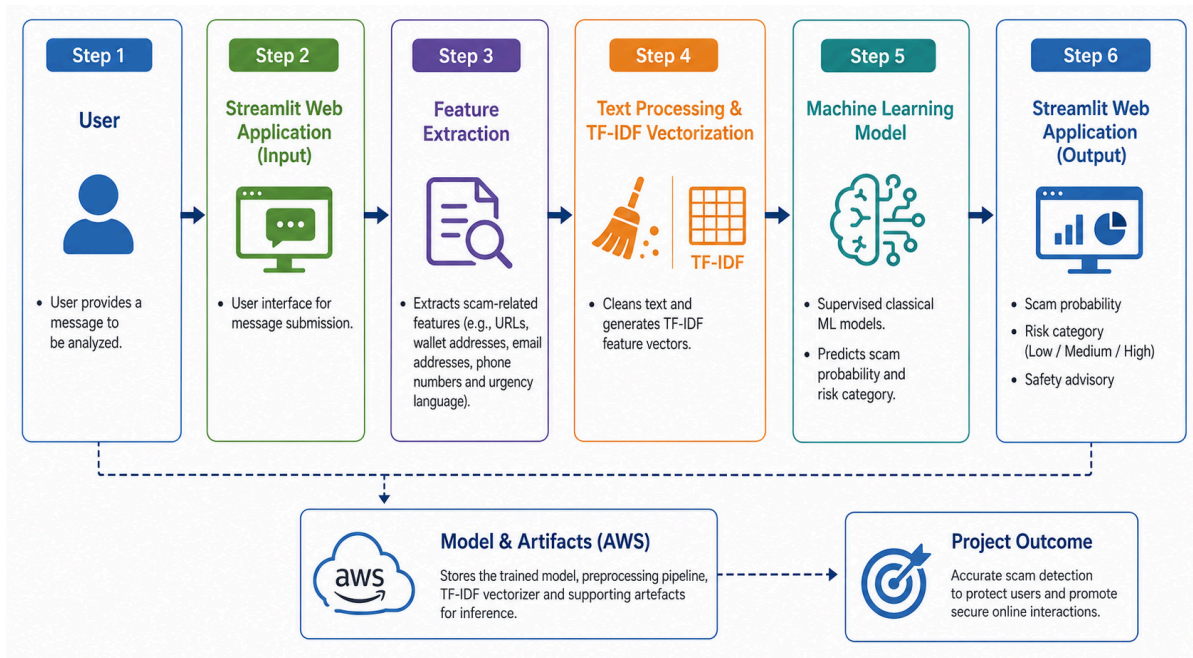
This prediction is based on patterns learned from the training data and should be interpreted as a risk assessment, not definitive proof that the message is fraudulent. The identified factors are commonly associated with cryptocurrency scams and therefore contributed to this risk score.

Recommendation:

- ➔ Do not transfer funds or disclose personal or financial information
- ➔ Note down any relevant information from the message for further verification
- ➔ Verify through official communication channels before taking any action.

→ If you suspect a scam, call the 24/7 ScamShield helpline at 1799 or report the incident through the relevant authorities.

System Architecture



System Architecture Description: The figure above illustrates the overall workflow of the proposed cryptocurrency scam detection system. This figure was created with the assistance of ChatGPT and subsequently reviewed and adapted by the project team.

6. AI Risk Assessment Identification – Please refer to the detailed AI System Risk and Impact Assessment from Page 6 onwards.

7. Team Members and Focus Areas

- Royston – Data exploratory analysis, model development, analysis, and evaluation.
- Hui Lin – MLOps and deployment design, and AI risk assessment.
- Both team members are jointly responsible for ensuring the AI system is developed ethically and responsibly. Key design decisions, model selection, deployment readiness, and risk mitigation measures are reviewed collaboratively before implementation to maintain accountability and reduce the likelihood of individual oversight.

AI System Risk and Impact Assessment

System Name: Crypto Text Scam Detector

Date of Assessment: 26 July 2026

Phase of Development: Concept / Proposal

I. System Context and Purpose

1. Primary objective and intended use case

- **Problem solved:** Cryptocurrency-related scams account for about 20% of total scam losses in Singapore⁵. Users often lack timely, reliable support to assess whether a crypto-related message is fraudulent before acting. Our system classifies crypto-related messages as scam or legitimate to give users early decision-support.
- **Intended users:** Members of the public who receive suspicious crypto-related messages via Telegram, WhatsApp, X/Twitter, Discord, Reddit, or Email
- **Core functionalities:**
 - Preprocesses pasted crypto-related text messages
 - Converts text to TF-IDF features vectors
 - Predicts scam likelihood using a trained classical ML model
 - Outputs the predicted risk level (Low/Medium/High), key contributing indicators, and a safety advisory
- **Limitations:**
 - Trained on a synthetic Kaggle dataset, not real-world scam messages
 - Proof-of-concept only, not validated for production deployment
 - Not a replacement for official verification (ScamShield, authorities)
 - Does not guarantee legitimacy even on low-risk predictions
 - The dataset is only in English and free of spelling errors, whereas scam messages often include multilingual text, deliberate misspellings, or unconventional grammar

2. Direct and indirect stakeholders

- **Direct:** End users pasting messages into the Streamlit app and the two developers
- **Indirect:** User's contacts, financial institutions, crypto platforms named in messages
- **Vulnerable/marginalised groups:** Elderly users, first-time crypto investors, and less tech-savvy individuals may be disproportionately affected as they are both more likely to be scam targets and more likely to trust an automated low-risk verdict

⁵ Singapore Police Force, 2025. *Annual Scam and Cybercrime Brief 2025*. [Online] Available at: <https://www.scamshield.gov.sg/archive/2025-scams-cybercrime-briefs/>

3. Deployment environment and operational context

- **Where it operates:** Cloud-hosted model, TF-IDF vectoriser, and preprocessing artifacts trained and stored on AWS to be presented via a Streamlit web application
 - **Interaction with other systems:** Streamlit web application, AWS SageMaker endpoint, and AWS S3 for model artifacts. No integration with external messaging platforms or third-party APIs.
 - **Expected scale:** Proof-of-concept / academic demo scale, not production traffic
 - **Human-in-the-loop:** The system is explicitly positioned as decision support, not an authoritative verdict. Users are advised to independently verify through official channels (e.g., ScamShield Helpline (1799) or the relevant authorities) before acting, especially on medium/high-risk predictions
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II. Data Considerations

4. Data used for training, validation, testing, and operation

- **Source:** Kaggle Crypto Scam Dataset. Publicly available, synthetically generated
- **Data type:** Structured tabular CSV. No personal or sensitive personal data. Synthetic, no real individuals, wallets, or platforms involved
- **Schema:**
 - id (unique identifier)
 - platform (Telegram, X/Twitter, SMS, Email, Discord, Reddit)
 - text (message content)
 - label (scam / legitimate)
- **Volume:** Target size 10,000 records
- **Data quality:** Synthetically generated for balanced, linguistically diverse coverage of scam typologies and legitimate crypto communication. Data quality will be assessed through exploratory data analysis (EDA), including checks for missing values, duplicate records, class balance, feature distributions, and platform representation

5. Potential biases or limitations in the dataset

- Synthetic data may over-represent certain scams and may not capture real-world scam language, regional communication styles, or newly emerging tactics
- No demographic data is present, so demographic bias is not directly assessable but linguistic/platform bias is a concern
- **Mitigation methods:** Use diverse scam categories and platforms, incorporate publicly available verified scam examples where appropriate, and periodically update the dataset with new scam patterns

6. Data privacy and security management

- **Training data:** Fully synthetic so there are no real individuals, wallet addresses, or platform accounts involved, minimising privacy risk
 - **Regulatory compliance:** Since user-submitted messages are not logged or persisted, Singapore PDPA laws for retention and storage obligations are largely avoided
 - **Operational data:**
 - User-submitted messages are not permanently stored
 - Inputs are processed only for inference
 - **Retention:** Zero retention policy. Messages are processed in-memory for inference only and are never written to disk or a database, not even temporarily within a session
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III. Model Design and Performance

1. Type of AI model

- **Approach:** Supervised, classical ML binary text classification
- **Candidate models:** Baseline heuristic/keyword rules, Logistic Regression, Naive Bayes, Support Vector Machine, Random Forest, Neural Network
- **Complexity/interpretability:** Deliberately kept classical and interpretable as a proof-of-concept. LLM-based approaches are noted as possible future work and improvement
- **Known limitations:** TF-IDF/classical models capture lexical patterns but have limited semantic understanding, making them vulnerable to ever-changing scam language

2. Accuracy, reliability, and robustness assessment

- **Evaluation metrics:** Accuracy, Confusion Matrix, Precision, Recall, F1 Score, ROC Curve/AUC, Precision-Recall Curve/AUC
- **Acceptable thresholds:** The goal is high recall with acceptable precision, since a false negative risks direct financial/identity harm. Recall $\geq 85\%$ (priority to minimise missed scams), precision $\geq 75\%$, F1-score $\geq 80\%$, ROC-AUC ≥ 0.85 .
- **Adversarial/OOD testing:** Not planned for at the proof-of-concept and proposal stage but for the final and production version, that can be done using a manually paraphrased scam test set to check pattern generalisation or perturbation testing (synonym swaps, typos) on existing test samples to measure robustness.
- **Drift detection/degradation:** Planned periodic performance monitoring, retraining with newer scam examples, and review after significant scam trends emerge
- **Fallback/human override:** System is decision-support only, every prediction is paired with a recommendation to verify through official channels. Models are version-controlled so a prior version can be restored if a new one underperforms.

3. Explainability and transparency

- **Explainability approach:** Output includes scam probability and the key suspicious indicators detected (e.g. urgency language, wallet address presence)
 - **Documentation:** System architecture is documented (User Input → Preprocessing → TF-IDF vectorisation → ML model → Risk Output)
 - **User awareness:** Users are clearly interacting with an AI decision-support tool while using the Streamlit interface
 - **Capabilities/limitations communicated to users:** Sample output includes a disclaimer along with recommendation, even low-risk predictions still carry a caution notice
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IV. Ethical and Societal Impact

1. Potential ethical implications

- **Bias/discrimination risk:** Synthetic training data may not generalise evenly across message, potentially producing uneven accuracy for certain scam types
- **Unfair/inequitable outcomes:** False negatives could disproportionately harm vulnerable users who are more likely to rely on and believe the tool's verdict
- **Autonomy/dignity:** Preserved as system is framed as decision-support, not an automated blocker of user action
- **Privacy/surveillance concerns:** Minimised by not storing user-submitted messages
- **Broader societal impact:** Positive potential to reduce scam losses and support public awareness. However there may be a risk of over-reliance, reducing users' own critical judgment over time

2. Fairness and non-discrimination

- **Fairness definition adopted:** The model will be considered fair if recall does not drop significantly for any single platform subgroup, since uneven detection across platforms would leave users on certain channels less protected than others
- **Bias mitigation methods:** Platform diversity included in dataset design to improve robustness at a proof-of-concept level. EDA planned to examine class imbalance, feature distribution, and platform representation
- **Ongoing monitoring:** Periodic evaluation and dataset updates as scam patterns evolve.
- **Redress/appeal process:** Recommend a feedback mechanism for users who feel a prediction was wrong

3. Potential harms (individual, group, societal, environmental)

Harm Area	Description	Likelihood	Severity	Existing Control	Residual Risk
Individual	False negatives may lead to users transferring funds to a scam believing it's legitimate (financial loss or identity theft)	Medium	Major	- Recall-optimised threshold - Confidence scores shown instead of hard binary decision	Medium
Individual	False positives for legitimate crypto communication flagged as scam may reduce user confidence / causing missed legitimate opportunity	Medium	Moderate	- Precision monitored alongside recall - Risk shown as probability, not absolute	Low - Medium
Individual	Over-reliance of the tool may lead to users treating low-risk as a guarantee and skips their own judgment	Medium - High	Major	- Disclaimers positioned explicitly as a decision-support tool. Encouraging verification via ScamShield (1799)	Medium
Organisational	Model degrades as scam language evolves, reducing trust in the tool over time	High (long-term)	Moderate	- Periodic performance monitoring aligns with retraining with new scam examples - Review after major scam trend shifts	Medium
Societal	Public may misunderstand the tool's scope and under-report to authorities, or spread false confidence in messages	Low - Medium	Moderate	- Explicit advisory to verify through official ScamShield or police channels regardless of prediction	Low

V. Governance and Accountability

1. Responsibility for governance, ethical oversight, and risk management

- **Roles:**
 - Royston – Data exploration, feature engineering, model development, hyperparameter tuning, model evaluation and performance analysis
 - Hui Lin – MLOps implementation, SageMaker deployment, experiment tracking, model version control, and AI risk assessment
- Both team members are jointly responsible for ensuring the AI system is developed ethically and responsibly. Key design decisions, model selection, deployment readiness, and risk mitigation measures are reviewed collaboratively before implementation to maintain accountability and reduce the likelihood of individual oversight

2. Regulatory and legal compliance

- **Specific AI regulations applied:**
 - Singapore's Model AI Governance Framework (IMDA/PDPC) given the local deployment context
 - Personal Data Protection Act (PDPA) principles by ensuring that no unnecessary personal data is collected, stored, or retained beyond what is required for system operation and any incidental data handling.
 - Compliance with AWS security best practices, including role-based access control (IAM), encrypted storage in Amazon S3 and role-based access control (IAM), and secure deployment using Amazon SageMaker

3. Incident and error handling

- **Version control:** All source code will be maintained using Git and GitHub. Trained models are version-controlled through SageMaker model artefacts and experiment tracking. If a newer model underperforms, the previous stable model can be restored quickly to minimise operational disruption
- **Error/bias reporting:** A feedback channel in future iterations for frontend development, such as allowing users to: report incorrect predictions, flag potentially biased or misleading results, and provide comments for continuous model improvement
- **Audit frequency:** Periodic monitoring Quarterly (every three months), with additional audits following major model updates, significant concept drift, or security incidents.

VI. Mitigation Strategies and Monitoring Plan

Risk Identified	Likelihood (Pre)	Severity (Pre)	Mitigation Strategy	Likelihood (Post)	Severity (Post)
Data Bias Synthetic data may not capture real-world scam diversity/regional styles/new tactics	Medium	Moderate	- Use diverse scam categories and platforms, incorporating public verified scam examples where appropriate - Periodically update dataset	Low	Moderate
Model Performance Risk False negatives (missed scams) & false positives (flagged legitimate messages)	Medium	Major	- Optimise classification threshold for higher recall, evaluating via accuracy, precision, recall, F1, ROC/PR-AUC - Display confidence scores instead of absolute decisions	Low - Medium	Moderate
Explainability Risk Users may not understand why a prediction was made, reducing trust	Medium	Moderate	- Show scam probability, not just binary label - Display key suspicious indicators - Clarify predictions are probabilistic	Low	Low
Human Over-reliance Users may treat prediction as infallible	Medium - High	Major	- Position app as decision-support only - Prominent disclaimers to encourage verification through official channels (ScamShield 1799)	Medium	Moderate
Concept Drift Scam language evolves, degrading model performance over time	High (long-term)	Moderate	- Monitor performance periodically - Retrain with newer scam examples - Review accuracy after major scam trend shifts	Medium	Moderate

Privacy and Security Users may paste wallet addresses, phone numbers, or personal info	Low	Moderate	- Do not permanently store user-submitted messages - Process inputs only for inference	Low	Low
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Monitoring Plan:

1. **Ongoing processes:** Periodic model performance review against defined metrics, monitoring for concept drift as scam tactics evolve, and model version control enabling rollback if a new version underperforms.
 2. **Frequency:** Quarterly (every 3 months), with additional audits conducted after any major model update, significant concept drift, or security incident.
 3. **Reporting/escalation:** Hui Lin (AI governance and MLOps lead) will be responsible for consolidating monitoring results and reporting identified risks to the project team
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VII. Review and Approval

Reviewed by:

- Royston Quek Jia Liang, Model Development [Reviewed 23/07/26]
- Ong Hui Lin, MLOps & Deployment [Reviewed 23/07/26]

Approved by:

- Course Tutor